

Path Forger

Project Introduction and Overview

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In one sentence: Path Forger turns resumes, transcripts, project notes, or guided student answers into an evidence-grounded career plan that users can inspect, edit, and act on.

The problem

Career guidance is easy to generate but difficult to trust. Background information is often scattered, goals are vague, and generic AI advice rarely shows which facts support a recommendation or what evidence a user should build next. Students, especially K-12 learners, may not have a traditional resume at all.

The solution

Path Forger creates a structured, reviewable profile before it recommends anything. The user confirms a career goal, the application gathers role evidence and calculates gaps, projects, and next actions, and a reasoner synthesizes the final plan. Warnings, evidence, provider details, latency, fallback events, and routing metadata remain visible.

Who it serves

- Students and early-career users turning scattered experiences into a practical direction.
- K-12 learners, parents, counselors, and teachers who need school-friendly exploration without requiring a resume.
- Career changers who want role-specific skill gaps, portfolio projects, and prioritized next actions.

What the current application demonstrates

- A focused page-based workflow: Overview, Student, Background, Profile, Goal, Results, and Review.
- Document or guided intake, editable extraction, sector/role/specialization selection, and dedicated results.
- Paths, strengths, ranked gaps, projects, next actions, evidence, caveats, warnings, and trace details.
- Gemini/OpenAI modes, local Qwen and HTTP-service contracts, hybrid fallbacks, and an offline demo mode.

How it works

- 1. Intake and extraction:** A BackgroundExtractor converts documents, pasted notes, guided answers, or K-12 context into a typed profile with evidence spans, confidence, and warnings.
- 2. Deterministic analysis:** Local tools gather role evidence, analyze fit and skill gaps, recommend portfolio projects, and rank next actions.
- 3. Reasoning and verification:** A CareerReasoner produces schema-bound plan data, while verification keeps unsupported claims, caveats, warnings, and trace fields visible.
- 4. Review and feedback:** The user inspects the plan on a dedicated Results page and can leave a browser-saved review for product improvement.

Technical design and AI usage

The React/Vite frontend calls a FastAPI backend with SQLite persistence. Agent orchestration depends on the BackgroundExtractor and CareerReasoner protocols. Gemini, OpenAI, Qwen, a local HTTP service, and the fake provider return compatible Pydantic structures, keeping the workflow testable and provider-neutral.

- Claude Code supported architecture planning and design/code review.
- OpenAI Codex supported focused implementation, debugging, tests, and documentation.
- Human review controlled product decisions, accepted changes, and determined what shipped.

Validation and responsible use

Backend tests cover schemas, tools, provider boundaries, fallbacks, verifier behavior, and API flow. The fake-provider evaluation uses five synthetic personas; each run is schema-valid with six evidence citations, while the ambiguous case produces a verifier warning. This validates workflow and trace behavior, not real-world model accuracy.

Demo focus

In about five minutes, the demo will show the journey from student/background context to an editable profile, a selected goal, an evidence-grounded plan, visible trace information, and a feedback path.